

Disproof of conjecture 00000001478

Polynomial images of cones and the tensor nuclear-norm ball

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Abstract

The nuclear-norm unit ball of complex $2 \times 2 \times 2$ tensors cannot be the image of a genuine cone under a polynomial map. A polynomial bounded on a cone is constant along each ray, and its value on every ray is its value at the common origin. The nuclear ball contains two points distinguished by a bounded coordinate, giving a contradiction. This rules out the conjectured spectrahedral-cone representation, including the stated 4×4 symmetric-matrix case.

1 The source claim and the precise negation

Both language versions of conjecture 00000001478 assert that the nuclear-norm unit ball in $\mathbb{C}^{2 \times 2 \times 2}$ is the polynomial image of an explicit spectrahedral cone described using 4×4 symmetric matrices, and that this representation gives SDP computability of the nuclear norm. The exact bilingual statement is included in `ORIGINAL.md`.

We disprove the necessary representation assertion. Thus no interpretation of the additional SDP consequence can make the stated conjunction true. We do not claim that the nuclear norm cannot be computed by some other SDP formulation.

The source abbreviates the nuclear norm as a *minimal sum of rank-one tensors*. We make its conventional scalar objective explicit. Let

$$V = \mathbb{C}^2, \quad \mathcal{T} = \mathbb{C}^{2 \times 2 \times 2},$$

with the usual Hermitian Euclidean norm on each factor V . For an arbitrary finite list $L = ((a_r, b_r, c_r))_r$ of triples in V^3 , put

$$A(L)_{ijk} = \sum_r (a_r)_i (b_r)_j (c_r)_k, \quad c(L) = \sum_r \|a_r\|_2 \|b_r\|_2 \|c_r\|_2.$$

Define the standard complex projective tensor norm and its closed unit ball by

$$D(T) = \{c(L) : A(L) = T\}, \quad \nu(T) = \inf D(T), \quad B = \{T : \nu(T) \leq 1\}.$$

The lists in $D(T)$ have unrestricted finite length. Every $D(T)$ is nonempty: the coordinate expansion

$$T = \sum_{i,j,k \in \{0,1\}} (T_{ijk} e_i) \otimes e_j \otimes e_k$$

gives a decomposition. All costs are nonnegative. Zero pure tensors can be deleted without changing either the sum or the cost: a pure tensor is zero exactly when a factor is zero. The formalization

proves these facts, so allowing zero factors does not change the attainable cost set for nonzero rank-one summands.

A cone here means a subset $C \subseteq \mathbb{R}^n$ containing 0 and satisfying $tx \in C$ whenever $x \in C$ and $t \geq 0$. A polynomial map $P : \mathbb{R}^n \rightarrow \mathcal{T}$ has real polynomials as the real and imaginary parts of all eight complex coordinates. Complex-coefficient polynomials on real parameters are included: their real and imaginary parts are real polynomials, as also proved in Lean.

Theorem 1. *For every natural number n , every such cone $C \subseteq \mathbb{R}^n$, and every polynomial map $P : \mathbb{R}^n \rightarrow \mathcal{T}$, one has $P(C) \neq B$.*

The quantifiers allow all finite parameter dimensions and all polynomial degrees. Convexity and a particular matrix presentation are not needed. This larger class contains every spectrahedral cone in the source claim.

2 A bounded coordinate and two exact points

Lemma 2. *For every tensor T and every coordinate i, j, k , $|T_{ijk}| \leq \nu(T)$. Consequently, $|\operatorname{Re} T_{000}| \leq 1$ for $T \in B$.*

Proof. For every factor triple,

$$|a_i b_j c_k| \leq \|a\|_2 \|b\|_2 \|c\|_2.$$

Applying the triangle inequality to any finite decomposition of T gives $|T_{ijk}| \leq c(L)$. This is a lower bound for every element of the nonempty set $D(T)$; hence it is at most its infimum. The real-part bound follows from $|\operatorname{Re} z| \leq |z|$. $\square$

Let $E = e_0 \otimes e_0 \otimes e_0$. The empty decomposition and nonnegative costs give $\nu(0) = 0$. The one-term decomposition of E has cost 1, while $E_{000} = 1$; Lemma 2 therefore gives $\nu(E) = 1$. In particular,

$$0, E \in B, \quad \operatorname{Re} 0_{000} = 0, \quad \operatorname{Re} E_{000} = 1.$$

These are exact equalities, not numerical estimates.

3 The polynomial obstruction

Lemma 3. *If a real polynomial p on $\mathbb{R}^n$ is bounded in absolute value on a cone C , then $p(x) = p(0)$ for every $x \in C$.*

Proof. Fix $x \in C$. Substitution $X_i = x_i t$ gives the univariate polynomial $q_x(t) = p(tx)$. Cone closure makes q_x bounded for every $t \geq 0$. A nonconstant real polynomial is unbounded in absolute value as $t \rightarrow +\infty$, by its leading term. Thus q_x is constant, and

$$p(x) = q_x(1) = q_x(0) = p(0).$$

No homogeneity of p , degree restriction, or interior point of C is assumed. $\square$

Proof of Theorem 1. Suppose $P(C) = B$. The real polynomial

$$p(x) = \operatorname{Re}(P(x)_{000})$$

is bounded by 1 in absolute value on C , by Lemma 2. Lemma 3 makes it equal to $p(0)$ everywhere on C . The image equality supplies $x_0, x_E \in C$ with $P(x_0) = 0$ and $P(x_E) = E$. Hence $p(x_0) = 0$ and $p(x_E) = 1$, a contradiction. $\square$

Only one coordinate is needed for the contradiction. More generally, coordinatewise application of Lemma 3 shows that a bounded polynomial image of a nonempty cone is a singleton.

4 The matrix condition and the word minimum

For a real linear pencil $L : \mathbb{R}^n \rightarrow \text{Mat}_4(\mathbb{R})$, define

$$C_L = \{x : L(x) \text{ is positive semidefinite}\}.$$

The PSD condition includes symmetry. The usual pencils with symmetric coefficient matrices are included in this definition. One has $0 \in C_L$, since $L(0) = 0$. If $x \in C_L$ and $t \geq 0$, then $L(tx) = tL(x)$ is symmetric and

$$v^\top L(tx)v = t v^\top L(x)v \geq 0 \quad \text{for all } v \in \mathbb{R}^4.$$

Thus C_L is a cone, and Theorem 1 rules out $P(C_L) = B$. Lean proves this closure directly from the genuine matrix PSD predicate. The general theorem also rules out any further generalized matrix description whose domain is an actual cone.

For completeness, the formalization separately treats the literal attained-minimum reading

$$B_{\min} = \{T : \text{some } r \in D(T) \text{ is least in } D(T) \text{ and } r \leq 1\}.$$

An attained least cost equals $\inf D(T)$, so $B_{\min} \subseteq B$. The costs 0 for the zero tensor and 1 for E are attained least costs, giving $0, E \in B_{\min}$. The proof of Theorem 1 works unchanged for every set B' with $\{0, E\} \subseteq B' \subseteq B$, and therefore for $B_{\min}$. No unproved global attainment theorem is used.

The argument concerns the cone and polynomial map stated in both languages. An affine section, normalization, or rational parametrization would require a different statement.

5 Formal verification and scope

The project uses Lean 4.19.0 and Mathlib v4.19.0, pinned by the supplied toolchain and dependency manifest. `NuclearNorm.lean` defines the cost set and proves the tensor bounds and exact points. `PolynomialCone.lean` proves the ray substitution, constancy, and complex-coefficient correspondence. `Solution.lean` combines them and proves the 4×4 PSD case. The main declaration is

`TLMC1478.conjecture00000001478_false.`

It is the negation of the existential representation by any genuine cone. The whole source conjunction implies this existential assertion, so its negation disproves the original conjecture.

The author build and strict source replays pass. `AuthorAudit.lean` checks all explicitly declared objects and results. The submission's additional verification records document the independent rebuild, the audit of all compiled mathematical declarations and their dependencies, and checks of the verification tools themselves. Reproduction instructions and precise results are in `README.md` and `VERIFICATION.md`. Proofs may depend only on the standard foundational axioms `propext`, `Classical.choice`, and `Quot.sound`. No admitted proof, custom axiom, or `native_decide` is used.

All mathematical computations are covered by the Lean proofs. No numerical optimization, external symbolic calculation, or auxiliary numerical program supplies a premise. The disproof needs neither all abstract norm axioms nor global attainment: the standard cost formula and the separately proved minimum-ball result suffice. Verification here is local; acceptance remains a maintainer decision.